

FPGA-Based Smart Hospital Emergency Response Dispatcher

User Manual

Board: Digilent Nexys A7-100T (Artix-7 XC7A100T)
Tool: Xilinx Vivado (VHDL-2008)
Course: 6th Semester Hardware Engineering — HSHL

1. System Overview

This system simulates a hospital emergency dispatch unit implemented entirely in hardware on an FPGA. A patient is registered via slide switches and a button press. The system then steps through a series of states — triage, resource allocation, dispatch, treatment monitoring, and release — all controlled manually via push buttons so each step is visible in real time on the board.

The system supports up to 4 simultaneous patients using the **park** feature, and individual patients can be released selectively using the **release mode**.

2. Board Layout

2.1 Push Buttons

Button	Location	Function
CPU_RESETN	Bottom left	Hard reset — clears everything
BTNL	Left	Register new patient
BTNC	Centre	Step FSM forward / confirm release
BTNR	Right	Emergency override (force Critical)
BTNU	Up	Park patient (keep resources busy, return to IDLE)
BTND	Down	Hold for release mode — then press BTNC to release

2.2 Slide Switches (SW7 – SW0)

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
ID3	ID2	ID1	ID0	Type1	Type0	Sev1	Sev0

Severity — SW[1:0] (rightmost)

SW1	SW0	Level
0	0	Normal

SW1	SW0	Level
0	1	Urgent
1	0	High
1	1	Critical

Type — SW[3:2] (middle)

SW3	SW2	Type
0	0	General
0	1	Accident
1	0	Stroke
1	1	Heart Attack

Patient ID — SW[15:12]: binary value 0–15 (0000 to 1111)

Note during release mode: While holding BTND, SW[1:0] selects the patient slot to release (00=slot 1, 01=slot 2, 10=slot 3, 11=slot 4). Set SW[1:0] before pressing BTNC.

2.3 LEDs

LED	Meaning
LED[0]	Room 1 occupied
LED[1]	Room 2 occupied
LED[2]	Room 3 occupied
LED[3]	Room 4 occupied
LED[4]	Doctor 1 occupied
LED[5]	Doctor 2 occupied
LED[6]	Doctor 3 occupied
LED[7]	Doctor 4 occupied
LED[8]	Equipment 1 in use
LED[9]	Equipment 2 in use
LED[10]	Equipment 3 in use
LED[11]	Equipment 4 in use
LED[12]	Critical patient flag
LED[13]	Treatment active
LED[14]	System fault

LED	Meaning
LED[15]	Alarm (blinks 2 Hz — latches until reset)

2.4 Seven-Segment Display

Eight digits read left to right:

AN7	AN6	AN5	AN4	AN3	AN2	AN1	AN0
FSM state	E / blank	d label	Doctor no.	r label	Room no.	Severity	Patient ID

FSM State codes (AN7):

Code	State
0	IDLE
1	RECEIVE PATIENT
2	TRIAGE
3	CHECK RESOURCES
4	ALLOCATE ROOM
5	ALLOCATE DOCTOR
6	ALLOCATE EQUIPMENT
7	DISPATCH TEAM
8	MONITOR TREATMENT
9	RELEASE RESOURCES
F	FAULT

Severity labels (AN1):

Symbol	Meaning
n	Normal
U	Urgent
H	High
C	Critical

AN6: Shows E when a critical patient is active or a fault has occurred. Blank otherwise.

3. FSM State Flow

IDLE → RECEIVE PATIENT → TRIAGE → CHECK RESOURCES
→ ALLOCATE ROOM → ALLOCATE DOCTOR → ALLOCATE EQUIPMENT
→ DISPATCH TEAM → MONITOR TREATMENT → RELEASE RESOURCES → IDLE

At any point: fault condition → FAULT (only CPU_RESETN recovers)
At DISPATCH: BTNU → park patient → IDLE (resources stay busy)

Each state requires a **BTNC** press to advance, except:

- **MONITOR TREATMENT** — advances automatically after a timer (1–4 seconds based on severity)
- **CHECK RESOURCES** — advances to FAULT automatically if no resources are free

4. Priority Scoring

The system calculates a priority score from severity and patient type:

Severity	Base Score
Normal	0
Urgent	1
High	2
Critical	3

Type	Boost
General	+0
Accident	+1
Stroke	+1
Heart Attack	+1

Maximum priority = 4 (Critical + any non-General type).

5. Demo Scenarios

Scenario 1 — Normal Full Flow

Purpose: Shows the complete patient lifecycle from registration to resource release.

Switch setup:

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
0	0	0	1	1	0	0	1

ID=1, Stroke, Urgent

Step	Action	Display	LEDs
1	CPU_RESETN	0 _ d 0 r 0 n 0	All off
2	BTNL	1 _ d 0 r 0 U 1	—
3	BTNC — TRIAGE	2 _ d 0 r 0 U 1	—
4	BTNC — CHECK	3 _ d 0 r 0 U 1	—
5	BTNC — ALLOC ROOM	4 _ d 0 r 1 U 1	LED[0] on
6	BTNC — ALLOC DOCTOR	5 _ d 1 r 1 U 1	LED[4] on
7	BTNC — ALLOC EQUIP	6 _ d 1 r 1 U 1	LED[8] on
8	BTNC — DISPATCH	7 _ d 1 r 1 U 1	LED[13] on
9	BTNC — MONITOR	8 _ d 1 r 1 U 1	wait ~2 seconds
10	Auto — RELEASE	9 _ d 1 r 1 U 1	LED[0,4,8] off
11	BTNC — IDLE	0 _ d 0 r 0 n 0	All off

Scenario 2 — Critical Heart Attack

Purpose: Shows maximum priority patient, critical flag, and longer treatment timer.

Switch setup:

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
0	1	1	1	1	1	1	1

ID=7, Heart Attack, Critical

Step	Action	Display	LEDs
1	CPU_RESETN	0 _ d 0 r 0 n 0	All off
2	BTNL	1 _ d 0 r 0 C 7	—
3	BTNC — TRIAGE	2 E d 0 r 0 C 7	LED[12] on
4	BTNC — CHECK	3 E d 0 r 0 C 7	—
5	BTNC — ALLOC ROOM	4 E d 0 r 1 C 7	LED[0] on
6	BTNC — ALLOC DOCTOR	5 E d 1 r 1 C 7	LED[4] on
7	BTNC — ALLOC EQUIP	6 E d 1 r 1 C 7	LED[8] on
8	BTNC — DISPATCH	7 E d 1 r 1 C 7	LED[13] on
9	BTNC — MONITOR	8 E d 1 r 1 C 7	wait ~4 seconds

Step	Action	Display	LEDs
10	Auto — RELEASE	9 E d 1 r 1 C 7	LED[0,4,8] off
11	BTNC — IDLE	0 _ d 0 r 0 n 0	LED[12] off

Scenario 3 — Emergency Override

Purpose: Shows BTNR forcing a Normal patient to Critical regardless of switches.

Switch setup: All switches down (Normal, General, ID=0)

Step	Action	Display	LEDs
1	CPU_RESETN	0 _ d 0 r 0 n 0	All off
2	BTNR (override latch)	no change	—
3	BTNL	1 _ d 0 r 0 n 0	—
4	BTNC — TRIAGE	2 E d 0 r 0 C 0	LED[12] on — forced Critical
5	BTNC×4 — to DISPATCH	7 E d 1 r 1 C 0	LED[0,4,8,13] on
6	BTNC — MONITOR	8 E d 1 r 1 C 0	wait ~4 seconds
7	Auto + BTNC	0 _ d 0 r 0 n 0	All off

Scenario 4 — Park 3 Patients (Multi-Resource Demo)

Purpose: Shows multiple patients occupying resources simultaneously using BTNU park.

Patient 1 — Urgent Stroke, ID=3

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
0	0	1	1	1	0	0	1
Step	Action	Display				LEDs	
1	CPU_RESETN	0 _ d 0 r 0 n 0				All off	
2	BTNL	1 _ d 0 r 0 U 3				—	
3	BTNC×6 — to DISPATCH	7 _ d 1 r 1 U 3				LED[0,4,8,13] on	
4	BTNU — park	0 _ d 0 r 0 n 0				LED[0,4,8] stay on	

Patient 2 — Critical Heart Attack, ID=7

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
0	1	1	1	1	1	1	1

Step	Action	Display	LEDs
5	BTNL	1 _ d 1 r 1 C 7	—
6	BTNC×6 — to DISPATCH	7 E d 2 r 2 C 7	LED[0,1,4,5,8,9,12,13] on
7	BTNU — park	0 _ d 0 r 0 n 0	LED[0,1,4,5,8,9,12] stay on

Patient 3 — High Accident, ID=A

SW15	SW14	SW13	SW12	SW3	SW2	SW1	SW0
1	0	1	0	0	1	1	0
Step	Action	Display				LEDs	
8	BTNL	1 _ d 2 r 2 H A				—	
9	BTNC×6 — to DISPATCH	7 _ d 3 r 3 H A				LED[0,1,2,4,5,6,8,9,10,13] on	
10	BTNU — park	0 _ d 0 r 0 n 0				LED[0,1,2,4,5,6,8,9,10] stay on	

Result: Rooms 1, 2, 3 — Doctors 1, 2, 3 — Equipment 1, 2, 3 all busy simultaneously.

Scenario 5 — Selective Release

Purpose: Shows releasing individual parked patients by slot without affecting others.

Continue from Scenario 4 (3 patients parked in slots 1, 2, 3).

Release Patient 2 — slot 2 (Room 2, Doctor 2, Equipment 2):

Step	Action	Display	LEDs
1	Set SW[1:0]=01 (slot 2)	—	—
2	Hold BTND	—	—
3	Press BTNC	unchanged	LED[1,5,9] go OFF
4	Release BTND	—	Room 2, Doctor 2, Equip 2 now free

Release Patient 1 — slot 1 (Room 1, Doctor 1, Equipment 1):

Step	Action	Display	LEDs
5	Set SW[1:0]=00 (slot 1)	—	—
6	Hold BTND , press BTNC	unchanged	LED[0,4,8] go OFF

Patient 3 remains: LED[2,6,10] still on — Room 3, Doctor 3, Equipment 3 still occupied.

Scenario 6 — Resource Exhaustion and FAULT

Purpose: Shows the system entering FAULT when no resources are available.

Park 4 patients using Scenario 4 steps (all 4 rooms full), then register a 5th:

Step	Action	Display	LEDs
1	Park 4 patients via BTNU	0 _ d 4 r 4 _ _	LED[0–3,4–7,8–11] all on
2	BTNL (5th patient)	1 _ ...	—
3	BTNC×3 — reach CHECK	3 _ ...	—
4	Auto — no rooms free	F E d 4 r 4 _ _	LED[14] on, LED[15] blinks
5	Press any button	no change	System locked
6	CPU_RESETN	0 _ d 0 r 0 n 0	All off

Scenario 7 — Full Reset from Any State

Purpose: Shows CPU_RESETN clears everything instantly from any state.

Step	Action	Result
1	Press CPU_RESETN at any point	Display resets to 0 _ d 0 r 0 n 0
2	—	All LEDs off
3	—	All resources freed
4	—	System ready for new patient

6. UART Output

Connect a serial terminal (PuTTY, Tera Term) to the Nexys A7 USB COM port:

- **Baud rate:** 115200
- **Format:** 8N1 (8 data bits, no parity, 1 stop bit)

On each DISPATCH the system sends: P + patient ID as ASCII hex + carriage return (\r)

Patient ID	UART Output
1	P1\r
7	P7\r
10 (A)	PA\r
15 (F)	PF\r

7. Quick Reference Card

Action	Button	Condition
Register patient	BTNL	FSM must be in IDLE
Step FSM	BTNC	Any state except MONITOR
Force Critical	BTNR	Press before BTNL
Park patient	BTNU	Only in DISPATCH state
Release slot	Hold BTND + press BTNC	Set SW[1:0] to slot first
Hard reset	CPU_RESETN	Any time
Display	Meaning	
AN7 = 0	System idle, ready	
AN7 = F	Fault — press CPU_RESETN	
AN6 = E	Critical patient or fault active	
AN1 = C	Critical severity	
LED[14] on	Fault detected	
LED[15] blinking	Alarm latched	

8. Common Issues

Symptom	Cause	Fix
BTNL has no effect	FSM not in IDLE	Check AN7 — step to IDLE or reset
Resources not releasing	Park mode used, not normal flow	Use BTND + BTNC to release slots
Display shows old patient after park	Expected — display cleared on BTNU	Confirmed working — resets on park
FAULT immediately at CHECK	All 4 rooms/doctors/equipment busy	Release a slot or CPU_RESETN
LED[15] blinking, nothing responds	Alarm latched in FAULT	Press CPU_RESETN
BTNC steps FSM while in release mode	BTND not held down properly	Hold BTND firmly before pressing BTNC